

SCHOOL
OF
INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAMME STRUCTURE

M.TECH. COMPUTER SCIENCE AND ENGINEERING

SPECIALIZATION : SOFTWARE ENGINEERING

Academic Batch 2026-2028 and Onwards

GAUTAM BUDDHA UNIVERSITY

GAUTAM BUDH NAGAR, GREATER NOIDA, UP, INDIA





SEMESTER I

S.No.	Course Code	Course Name	L	T	P	Credits	Type
1	CS221	Advanced Data Base Management System	1	0	0	1	MC
2	CS224	Advanced Data Structure and Algorithms	1	1	0	2	MC
3	CS225	Operating Systems in RT	1	0	0	1	MC
4	CS227	Python Programming	1	0	0	1	MC
5	CS228	Entity and Normalization	1	0	0	1	MC
6	CS231	Advanced Data Base Management System Lab	0	0	4	2	MC
7	CS233	Python Programming Lab	0	0	1	1	MC
Total Hours and Credits			15	1	5	20	

SEMESTER II

S.No.	Course Code	Course Name	L	T	P	Credits	Type
1	CS222	Advanced Software Engineering	1	0	0	1	MC
2	CS224	Deep Learning	1	0	0	1	MC
3	CS226	DeepOps Engineering	1	0	0	1	MC
4	CS228	Blockchain Technology	1	1	0	2	MC
5	CS230	Empowering the Edge	1	0	0	1	MC
6	CS232	Advanced Software Engineering Lab	0	0	4	2	MC
7	CS234	Deep Learning Lab	0	0	4	2	MC
Total Hours and Credits			15	1	8	20	

ELECTIVES FROM DCE

S.No.	Course Code	Course Name	L	T	P	Credits	Type
1	CS330	Software Reliability Engineering	3	0	0	3	MC
2	CS332	Software Quality Assurance	3	0	0	3	MC
3	CS334	Software Project Management	3	0	0	3	MC
4	CS336	Open Source Software Engineering	3	1	0	3	MC
5	CS338	Parallel and High Performance Computing	3	0	0	3	MC

SEMESTER III

S.No.	Course Code	Course Name	L	T	P	Credits	Type
1	CS621	Cloud and Edge Computing	3	0	0	3	MC
2	CS623	AI Methods for Software Engineering	3	0	0	3	MC
3	CS624	AI Methods for Software Engineering	3	0	0	3	MC
4	CS626	Programme Elective-III	3	0	0	3	MC
5	CS628	AI-Assisted Software Engineering Lab	0	0	4	2	MC
6	CS630	Dissertation Part I	0	0	0	0	MC
Total Hours and Credits			12	0	4	20	

SEMESTER IV

S.No.	Course Code	Course Name	L	T	P	Credits	Type
1	CS692	Dissertation Part-II	0	0	40	20	MC
Total Hours and Credits			0	0	40	20	

ELECTIVE BASKET-III/II (SELECT ANY TWO)

S.No.	Course Code	Course Name	L	T	P	Credits	Type
1	CS631	Advanced Software Architecture and Design	3	0	0	3	MC
2	CS633	Cloud-Native and Microservices Engineering	3	0	0	3	MC
3	CS635	Advanced Software Testing	3	0	0	3	MC
4	CS637	Software Measurement and Estimation	3	0	0	3	MC
5	CS639	Secure Software Engineering and DevSecOps	3	0	0	3	MC
6	CS641	Empirical Software Engineering	3	0	0	3	MC
7	CS643	Software Engineering Analytics	3	0	0	3	MC
8	CS645	Software Product Management and Requirements Analytics	3	0	0	3	MC
9	CS647	Distributed Systems and Middleware	3	0	0	3	MC
10	CS649	Software Maintenance	3	0	0	3	MC